5.6.2 The Weibull Distribution

The Weibull distribution is widely used in engineering practice due to its versatility. It was originally proposed for the interpretation of fatigue data, but now its use has been extended to many other engineering problems. In particular, it is widely used in the field of life phenomena as the distribution of the lifetime of some object, especially when the "weakest link" model is appropriate for the object. That is, consider an object consisting of many parts, and suppose that the object experiences death (failure) when any of its parts fail. It has been shown (both theoretically and empirically) that under these conditions a Weibull distribution provides a close approximation to the distribution of the lifetime of the item.

The Weibull distribution function has the form

$$F(x) = \begin{cases} 0 & x \le \nu \\ 1 - \exp\left\{-\left(\dfrac{x - \nu}{\alpha}\right)^{\beta}\right\} & x > \nu \end{cases} \tag{6.2}$$

A random variable whose cumulative distribution function is given by Equation (6.2) is said to be a *Weibull random variable* with parameters ν, α, and β. Differentiation yields the density:

$$f(x) = \begin{cases} 0 & x \le \nu \\ \dfrac{\beta}{\alpha}\left(\dfrac{x - \nu}{\alpha}\right)^{\beta-1} \exp\left\{-\left(\dfrac{x - \nu}{\alpha}\right)^{\beta}\right\} & x > \nu \end{cases}$$

5.6.3 The Cauchy Distribution

A random variable is said to have a Cauchy distribution with parameter $\theta, -\infty < \theta < \infty$, if its density is given by

$$f(x) = \frac{1}{\pi}\frac{1}{1 + (x - \theta)^2} \qquad -\infty < x < \infty$$

EXAMPLE 6b

Suppose that a narrow-beam flashlight is spun around its center, which is located a unit distance from the x-axis. (See Figure 5.7.) Consider the point X at which the beam intersects the x-axis when the flashlight has stopped spinning. (If the beam is not pointing toward the x-axis, repeat the experiment.)

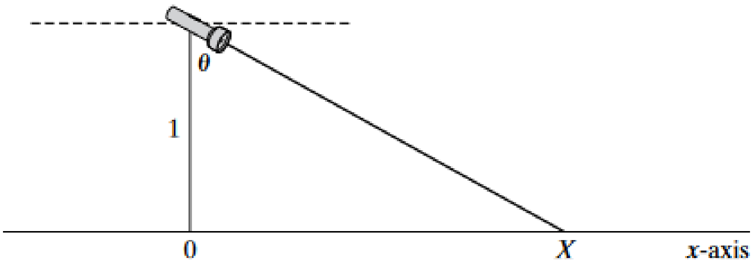

FIGURE 5.7

As indicated in Figure 5.7, the point X is determined by the angle θ between the flashlight and the y-axis, which, from the physical situation, appears to be uniformly distributed between $-\pi/2$ and $\pi/2$. The distribution function of X is thus given by

$$\begin{aligned} F(x) &= P\{X \leq x\} \\ &= P\{\tan\theta \leq x\} \\ &= P\{\theta \leq \tan^{-1} x\} \\ &= \frac{1}{2} + \frac{1}{\pi}\tan^{-1} x \end{aligned}$$

where the last equality follows since θ, being uniform over $(-\pi/2, \pi/2)$, has distribution

$$P\{\theta \leq a\} = \frac{a - (-\pi/2)}{\pi} = \frac{1}{2} + \frac{a}{\pi} \qquad -\frac{\pi}{2} < a < \frac{\pi}{2}$$

Hence, the density function of X is given by

$$f(x) = \frac{d}{dx}F(x) = \frac{1}{\pi(1 + x^2)} \qquad -\infty < x < \infty$$

and we see that X has the Cauchy distribution.† ■